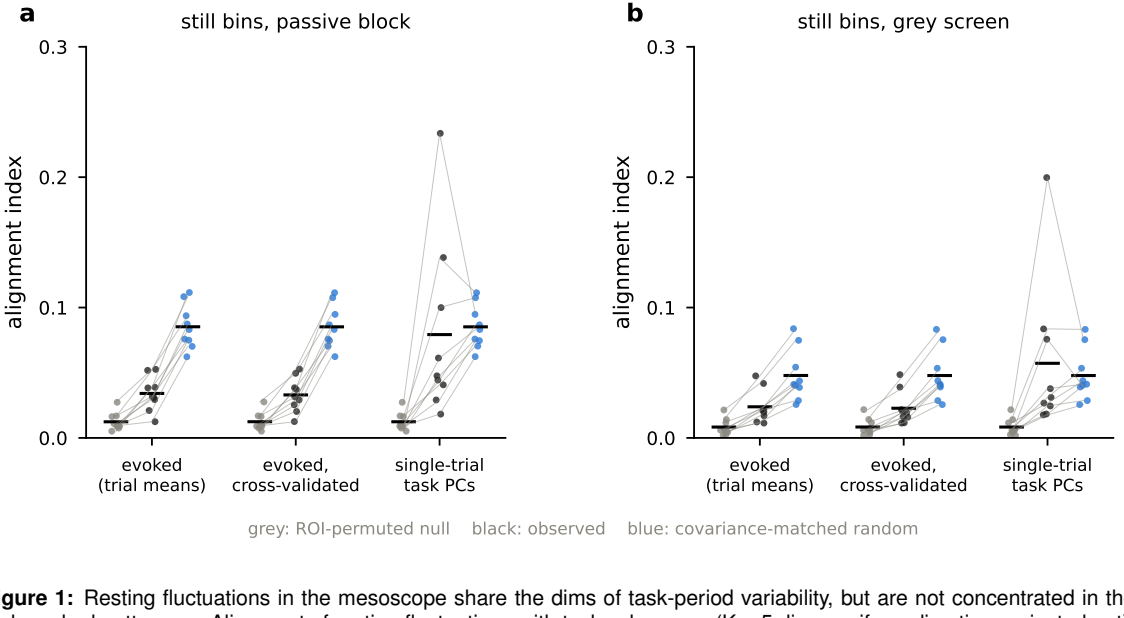

Figure 1: Resting fluctuations in the mesoscope share the dims of task-period variability, but are not concentrated in the task-evoked patterns. **a** Alignment of resting fluctuations with task subspaces (K = 5 dims; uniform direction projected out), per mouse (dots; bars, means over 9 mice; lines connect a mouse). Rest: still 1 s bins of the passive block (grey screen and receptive-field mapping), z-scored deconvolved activity of all ROIs, centred. Alignment index: rest variance in the subspace relative to the most any 5-dim subspace captures. Task subspaces: top dims of 16 trial-averaged evoked patterns (stim, choice, feedback, post-feedback windows minus pre-stimulus; side $\times$ outcome); the same restricted to dims shared by odd and even trials; top PCs of single-trial evoked activity (odd trials). Nulls: ROI permutation of the task patterns (grey) and random subspaces drawn from the covariance of all recording bins (blue). **b** The same for still bins of the grey screen alone (34 sessions).